

A Proposed Model for Integrating AI into Product Design Thinking Education

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Abstract— As AI reshapes the educational landscape, rethinking instructional strategies to foster Design Thinking in engineering is no longer optional—it is essential. This paper proposes a framework integrating AI into the design workflow, validated through a comparative study on a computer mouse design project. The results showed a profound difference: the group working with the support of AI outperformed the traditional control group significantly, as they showed greater creativity, speed of ideas, and the total amount of designs produced. Most importantly, the application of AI acted as a catalyst here, widening the scope of the designs, nudging the students in the direction of varied solution pathways. Feedback received from the following 60 students of the next cohort further supported that the use of the tools of AI brought a dramatic positive change in the learning motivation of the students.

Keywords— *Artificial intelligence, design thinking, engineering education, teaching methods, product design, computer mouse.*

I. INTRODUCTION

The advent of the Fourth Industrial Revolution has rendered the integration of Artificial Intelligence (AI) into engineering education not merely critical, but inevitable. AI has evolved from being a simple tool. Now, it has become an interactive assistant, allowing students to be more creative, ask better questions, and provide solutions to challenging problems [1][10][12][13][14].

In the field of industrial design, AI applications such as ChatGPT, Midjourney, and SolidWorks AI Add-in applications can be applied at each stage of product development, from conceptual stages to the creation of detailed prototypes. Using ChatGPT can analyze and structure consumer insights, formulate smart questions, and make sense of qualitative data in order to inform enhancements. Midjourney allows rapid exploration by generating images from text, while SolidWorks AI assists in drafting, enabling the conversion of CADs into accurate 2D drawings. Consequently, these tools not only enhance efficiency but also expand the design space via a visual, intelligent, and data-driven approach [2][11][14].

However, technical proficiency in these tools is insufficient; engineering students must also possess a grounded understanding of how to sustain and optimize design decisions through data and User Experience (UX). For ergonomic products like computer mice, Design Thinking (DT) serves as

the primary methodology for achieving optimal UX [3]. Yet, students often struggle to transition from the linear, logical reasoning typical of engineering to the divergent, creative thinking required in design [12][13]. Furthermore, without proper pedagogical scaffolding, AI integration risks fostering "solution duplication" and intellectual dependency, potentially eroding critical thinking skills. To address these challenges, this study explores two key questions:

1. How can AI be integrated into the Design Thinking framework to maximize support while maintaining the teacher's leadership role?

2. How can the development of design thinking skills be accurately assessed within this AI-augmented learning environment?

II. LITERATURE REVIEW

A. Design Thinking: The Foundational Pedagogic Framework

Design Thinking (DT) is firmly rooted in the realm of user-focused cognitive thinking, with an iterative process that involves five essential cycles: empathize, define, ideate, prototype, and test [3]. For engineering education, Design Thinking has been considered an essential teaching tool for instilling primary attributes in graduates, namely the skills to resolve complex problems, higher-order thinking, and working with multiple disciplines to further the goals of would-be engineers [15]. There has been a copious amount of literature to validate that Design Thinking helps in integrating knowledge with in-depth understanding of users to develop feasible and more innovative solutions than existing methods of instruction focusing on theoretical learning and mathematical calculations [4][16][17].

B. The Evolving Role of AI in the Design Lifecycle

The rising trend of various AI ecosystems, such as Large Language Models (LLMs), Text-to-Image models, and Generative Design tools, has changed the role of AI from being a mere automation aid to being a "co-create partner" in the designing process. The roles allotted to AI currently range from helping in concept variations to suggesting geometric form optimization via generative and topology optimization to simulating initial user interaction to assisting in writing technical

documentation. Many authors feel that the coming age of Gen AI has great transformative powers in terms of engineering education, ranging from the level of students to the level of the program [5][10][20][21][22].

C. Principles for Scaffolded AI Integration in Curriculum Design

In terms of designing learning, the integration of AI should not involve any unbiased “full tool unlock” initially but must instead adopt the model of phases in integrating AI [17][18][19]:

- Pre-AI Phase: The students are encouraged to work on initial ideas by themselves to help them work independently.

- Co-use Phase: The instruction emphasizes the importance of mastering the art of prompt engineering, in-depth AI analysis, and the capability to evaluate different solutions designed by AI tools.

- Post-AI Phase: The students would be asked to evaluate, improve, and heavily edit the AI results, in addition to maintaining an “AI-contribution” log to facilitate transparency in terms of technology utilization.

There are current best practices in teaching that involve the inclusion of skills such as prompt engineering, output analysis, and developing metacognition (urging students to ask: “What did the AI tool enhance, and what was the unique contribution of the human?”). The proposed action plan allows AI to be considered an augmentation factor rather than a replacement for human intelligence [6].

D. Reframing Faculty Roles and Learning Activities

Faculty in the AI-integrated model: The function of the school faculty in the AI-integrated model shifts to the “architect of learning experiences and critical reflections” from being the “transmitter of knowledge.” The educator must plan focused micro-tasks to direct the student on the purposeful application of AI, complete in-depth rubrics on process and final outputs, and offer explicit instruction on ethics related to technology, such as AI ethics and copyright sourcing [17][18]. Recommendations from prominent technology instruction institutes firmly suggest that questioning “how AI helped in that respective step” after completing each exercise with regularity is imperative to facilitate “reflective practice” [7][23].

E. Assessing Design Thinking Competence in an AI-Augmented Environment

** The Assessment Challenge:*

The involvement of AI in the process of creativity presents an evaluation dilemma in terms of separating human work from AI results. This significantly affects the aspect of academic integrity in grades, especially in terms of whether good grades in projects represent the actual level of creativity in students or rather the level of competency in the process referred to as “task optimization” or even “academic integrity in AI-generated

results.” The current state of literature emphasizes the importance of revising policies to suit the current state of affairs [8][23].

**Proposed Assessment Solutions:*

Research and learning practice provide different valid methods for evaluation:

- AI Contribution Statement: Requiring students to provide a well-articulated statement of the specifics of AI usage, where, when, and why it was done, and how it was transformed further. Such methods would help teachers to delineate clearly AI-recommended aspects from the innovation put together by students in AI tools. Templates are being rapidly designed for the preparation of such AI contribution statements [9].

- Multi-Criteria Rubrics (Process + Product) – Using rubrics designed to assess the gamut of criteria: intensity of user empathy, novelty of autonomous ideas, level of refinement done to AI output, efficacy of defending one’s design judgments, and proper utilization of AI. Such rubrics would involve individual evaluation criteria for “Original Idea Generation” and “Post-AI Refinement” to make it more transparent in nature [4].

- Design Process Analytics: The utilization of digital records, version histories in files, prompt history, and process analysis dashboards to measure the number of iterations, time span between versions, and group contributions to the process. This measurement approach shifts the evaluation criteria for process improvement not only to the final outcome but to the entire process of design behavior [6].

**Academic Integrity and Ethical:*

Apart from technology-enabled evaluation methods, policies related to Academic Integrity must be framed to emphasize best practices concerning the application of GenAIs to the teaching realm. These policies should state requirements to disclose AI tools applied to assignments, avoid misrepresenting AI-developed work as original, and introduce users to the risks associated with copyright violations and biases in models being applied to assignments. The latest norms issued by prominent institutions can offer an initial approach to implementing GenAIs in the engineering field for ethical purposes [8].

III. RESEARCH METHODOLOGY AND PEDAGOGICAL MODEL DESIGN

This section details the rationale for the design of the instructional model, the implementation plan of the empirical study, and finally the data collection and analysis procedure.

A. Designing the AI-Integrated Pedagogical Model in Design Thinking

The proposed work brings out a very organized teaching model in which AI is integrated at every stage of the conventional Stanford Design Thinking model. Notably, the proposed model has been designed with the cognitive scaffolding principle in mind, where AI is used to enhance the capability of students only after they complete an initial stage of solo ideation and critical analysis, thereby directly tackling the aspect of overdependence on technology and emphasizing solo critical judgment in the process [17][18]. AI is thereby transformed to play the role of an engaged learning partner in

which the capabilities of the machine and human are enhanced by qualitatively improving the learner's own capability for systematic observation, in-depth analysis, and final synthesis.

The detail-to-detail implementation of the proposed methodology is described in the table I below.

TABLE I. THE STRATEGICALLY SCAFFOLDED AI INTEGRATION MODEL

Phase	Core Cognitive Objective	Essential Student Actions	Strategic AI Augmentation	Instructor's Transformative Role
Empathize	- Gain insights into user needs, behavior, and technical limits (e.g., mouse ergonomics).	- Qualitative surveys and early interviews; - Behavioral observations; - User archetype definition (e.g., power vs. casual).	- Clustering + sentiment analysis of feedback; - Simulated AI personas for user types.	- Ensure well-designed, unbiased surveys; validate user sample representativeness; integrate technical knowledge with human-centered empathy.
Define	- Develop concise, actionable problem definitions and non-negotiable design requirements.	- Decompose user needs into constraints; - Define core problem (POV); - Set technical and ergonomic specs.	- Cluster technical challenges and pain points; - Convert qualitative data into testable design goals.	- Critique AI-human integration; - Pose dialectical questions; - Test underlying assumptions.
Ideate	- Maximize diversity and range of creative ideas within given boundaries.	- Group brainstorming sessions; - Concept sketch generation; - Structured analysis and concept selection.	- LLMs/GenAI for detailed design permutations; - Text-to-image for fast visual prototyping.	- Compare human vs. AI-generated outputs; - Require rationale for AI reliance; - Teach prompt engineering techniques.
Prototype	- Cast the chosen concepts into functional physical or virtual prototypes to validate.	- 3D CAD modeling; - Prototyping or simulation; - Tactile and ergonomic testing.	- AI-driven design optimization (weight, strength); - Auto-generation of engineering drawings from CAD.	- Ensure CAD-environment compliance - Validate technical and aesthetic standards - Evaluate manufacturability in real conditions
Test & Feedback	- Test highly stringent prototypes, accumulate empirical data, and perform design iteration.	- Real-world prototype deployment; - Collect quantitative + qualitative feedback; - Identify failure modes and recommend fixes.	- Automated AI-driven feedback surveys; - Advanced analytics (clustering, sentiment); - Data-based improvement proposals.	- Maintain testing objectivity and integrity; - Teach critical interpretation of AI outputs; - Run inter-group critique and reflection sessions.

B. Pedagogical Approach with Learning-Augmentation

The approach to instruction has been carefully planned to enhance critical autonomy and technological capability. The key ingredients ensuring AI acts only as an aid to learning and not as a replacement to human effort in the process are:

- Building AI Literary Knowledge: Mastery in AI Prompt Engineering The initial phase of the module has assigned

sessions to learn AI Prompt Engineering. The learning involves not only understanding basic queries on AI but mastering the art of designing structured queries, adjusting model parameters—Variability and Creative Temperature, and structuring models to help them evaluate and identify the best possible AI outcome. This helps to move the students from being passive to being active and collaborative AI participants.

- Aiding Critical Synthesis Approach to AI: Comparative ReflectionThe students are expected to go through a process of critical synthesis to fill the gap between human insights and AI outputs. Following the production of variations in AI, students are expected to write mini-reports on the variations in the design process: (1) The First Human Insight -> (2) AI Prompt -> (3) AI Development -> (4) Humans' Subsequent Improvement

- Personalized Scaffolding through AI-Assisted Feedback: The instructional team (lecturers/TAs) strategically utilizes AI to provide personalized scaffolding. Based on individual student progress or documented challenges, AI models can be employed to generate tailored hints or prompt suggestions designed to offer specific conceptual variations, effectively providing just-in-time assistance to students who may be struggling with early stage ideation. This ensures support is targeted and efficient

- Mandating Transparency and Accountability: The AI Contribution Statement To uphold academic integrity and facilitate accurate assessment, students are obligated to submit an AI Contribution Statement with all major deliverables. This formal declaration requires explicit documentation detailing when, how, and for what purpose AI was utilized, along with the rationale for selecting specific machine outputs. This compulsory disclosure enables the instructor to effectively disaggregate the learner's independent cognitive contribution from the technological support, directly supporting the objective assessment of design thinking competence.

C. Experimental Design and Investigation Groups

* Phased Research Design and Participant Allocation

Overall, it is pertinent to highlight that a two-phase research strategy was applied in this study to assess the efficacy of the designed AI integrated curriculum.

Phase 1: Pilot Validation (Semester 2024.1)

A standard class at Hanoi University of Science and Technology (HUST) is expected to hold about 60 students, divided equally into six groups. Considering the fact that each group will focus on a different subject associated with their academic discipline, it is expected that there shall be some natural diversity in their output.

This particular study tested the method on the 2024.1 semester, undergoing one complete cycle. This experiment was conducted on 20 students (N = 20) registered for the Art & Industrial Design class, who were allocated equally into two different groups, as discussed below.

Group A (Control Group): Participants strictly followed “Traditional Design “Methodology without using “AI as a cognitive or generative assistant” (Using standard CAD and modeling software only).

Group B (Intervention Group): Participants fully engaged in “AI-Integrated Learning Pedagogical Model” as defined in Section 3.1 of this research, actively involving “AI as a cognitive or generative assistant”.

Both groups were assigned an identical core assignment that involved research & development of a computer mouse design. This was an all-encompassing assignment that involved all of “Design Thinking”: User surveys, Pain identification, Idea generation, 3D modeling & testing, analysis & revisions to designs.

At the end of 2024.1 semester, preliminary analysis showed significant efficacy in learning outcome efficacy in terms of “AI integration” in Group B, thus prompting continued research in “Phase 2”.

Phase 2: Scaled Deployment & Focused Training (Semester 2024.2)

Building on the successful pilot study in Phase 1, the full cohort of the Industrial Art & Design course (Semester 2024.2) underwent comprehensive AI literacy training. This tutorial series focused specifically on the professional application of generative AI tools, including Midjourney and ChatGPT, within design workflows.

*Data Acquisition & Analytical Strategy:

Baseline Assessment: Prior to the project, a pre-test was administered to all Phase 1 participants to gauge their pre-existing AI literacy. This quantitative analysis focused on their proficiency with specialized tools (Midjourney, ChatGPT, and SolidWorks AI add-ins). The results confirmed a uniform lack of prior experience across the cohort, establishing a valid baseline for the experiment.

*Multi-Dimensional Evaluation Criteria :

Key dependent variables were assessed across three domains:

Cognitive & Application Skills: Design thinking, ideation (generation, screening, validation), and complex problem-solving.

Engagement & Proactivity: Collaborative dynamics, self-directed research, and critical questioning.

Attitudinal & Motivational Outcomes: Learning motivation, enthusiasm, and the adoption of an innovator’s mindset.

* Statistical Analysis:

Data were processed using Microsoft Excel and SPSS (Statistical Package for the Social Sciences). We employed appropriate inferential statistical tests to determine significant performance differences between the Control and Intervention groups.

IV. RESULTS AND DISCUSSION

A. Evaluating the Efficacy of AI Integration in Instruction

The effectiveness of the AI-integrated pedagogical model was systematically assessed through an empirical study comparing two student groups across the five stages of the Design Thinking (DT) process. The comparative findings detailing the behavioral and outcome differences between the Control Group (A) and the Intervention Group (B) are summarized in Table II.

TABLE II: COMPARISON RESULT BETWEEN GROUP A (TRADITION) & B (AI INTEGRATED) ON DT PHASES

ID	DT Phase	Group A (Traditional)	Group B (AI-Integrated)	Analytical Observations
1	Empathize	- Students relied on acquiring user feedback only on the Logitech website to identify existing user problems.	- AI was used for sophisticated filtering & grouping of existing user problems aggregated from different sources ONLINE	- Faster information retrieval (Group B) -Improved relevance and task focus -Identified AI skill gap in survey data processing
2	Define	- Subjective problem framing based on interpreted user input.	- AI quantified issues and justified core problem framing.	- AI aided justification and feasibility feedback, directing student focus.
3	Ideate	- Conventional Brainstorming and SCAMPER yielded limited initial ideas but led to focused tech and design improvements.	- AI provided core creative ideas, followed by Midjourney usage for fast visual processing & design concept creation	- Prolific and diversified ideation via AI; - Limited feasibility of outputs; - Need for explicit engineering constraints.
4	Prototype	- Students used standard AutoCAD and SolidWorks to create 3D CAD models and printed prototypes.	- Students used SolidWorks AI Add-ins for automatic 2D technical drawings of the mouse design.	- AI improved drafting precision, standardization, and reduced time through accurate dimensioning.
5	Test	- Teams had problems and limitations in their work due to the unavailability of prototypes for real users to test.	- Created a virtual model, with the help of AI, automated feedback survey creation, followed by initial analysis of feedback data	- AI supported virtual model development; however, the validity of simulation feedback remains uncertain and requires further study.

A comprehensive comparison of the effectiveness of learning was made between the Control Group (A) and the Intervention Group (B) with respect to the time taken to complete various tasks and the scores on the final assessment.

The timing was conducted with the help of an accurate stopwatch method for objective collection of data in the class. The findings are presented in Table III, which indicates the differences between the two groups.

TABLE III : COMPARATIVE LEARNING EFFECTIVENESS METRICS BETWEEN GROUPS A AND B

ID	Criterion	Group A (Traditional)	Group B (AI-Integrated)	Analytical Observations
1	Time for Empathize Phase (minutes)	120	36	- Group B spent less time on the Empathize Phase, which indicates the efficiency of AI.
2	Time for Define Phase (minutes)	62	28	- Group B had a great time advantage, showing that using AI makes problem definition faster.
3	Number of Ideas Generated	10	24	- Group A had difficulty producing ideas beyond one per student, while the other group realized a 240% boost in ideas generated.
4	Ideation Development Time (minutes)	156	20	- Group B reached 87% reduction in time, due to the rapid development of ideas by the students utilizing the AI system.
5	Creativity Assessment Score (Scale: 100)	75	86	- Group B scored highly on the creativity index, with greater diversity in the final product offering ideas, compared to the

6	Final Course Score (Scale: 100)	78	92	- Group B scored much higher on the average final course score, due to their improvements on the component exercises.
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The data indicates that the integrated AI model significantly accelerates the divergent phases of Design Thinking (Empathize, Define, and Ideate), resulting in superior final design outcomes. However, the results also underscore the concern raised in 'Criterion 4': the rapid speed of AI generation

may inadvertently hinder the development of independent creative thinking by fostering cognitive dependency. To address this, we established a creativity assessment framework designed to evaluate students' intrinsic performance distinct from AI assistance, as detailed in Table IV.

TABLE IV. ASSESSMENT FRAMEWORK FOR STUDENT CREATIVITY (INDEPENDENT OF AI)

Criteria	Percentage	Objective	Group A (Traditional)	Group B (AI-Integrated)
Originality (Pre-AI Creativity)	25%	Ensures the initial concept originates from the student.	87.8	87.3
Depth & Divergent Thinking	20%	Measures the expansion of the solution space.	87.6	87.8
Transformative AI Use	20%	Evaluates the ability to refine and transform AI outputs.	25.6	83.6
Human Reasoning & Justification	15%	Ensures design decisions are grounded in logic, not just AI suggestions.	86.6	83.0
Iterative Improvement	10%	Measures progress through design iterations.	87.8	86.8
User-Centered Creativity	10%	Ensures creativity is aligned with user insights.	86.7	87.3



Fig. 1. Traditional Ideation Outputs from Control Group A.

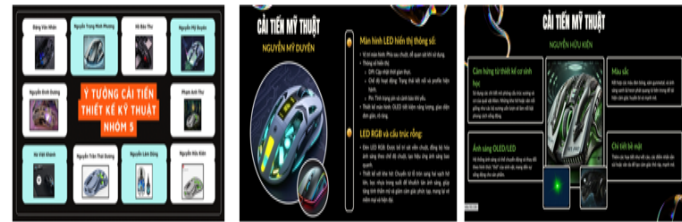


Fig. 2. AI-Augmented Ideation Outputs from Intervention Group B.

Figure 2 (Group B with AI support) shows consistently better results than Figure 1 (Group A using manual methods) across the main evaluation criteria. Group B obtained a higher creativity score (86 vs. 75), indicating a higher level of idea diversity in the final design outcomes. Regarding ideation time, Group B completed the ideation phase in approximately 20 min, whereas Group A required 156 min, demonstrating the efficiency of AI-based tools in facilitating the exploration and expansion of the design space. From a technical standpoint, the

designs produced by Group B exhibit greater detail and accuracy, achieved through the use of AI-based add-ins for automated dimensioning and annotation, resulting in more complete and consistent 2D drawings.

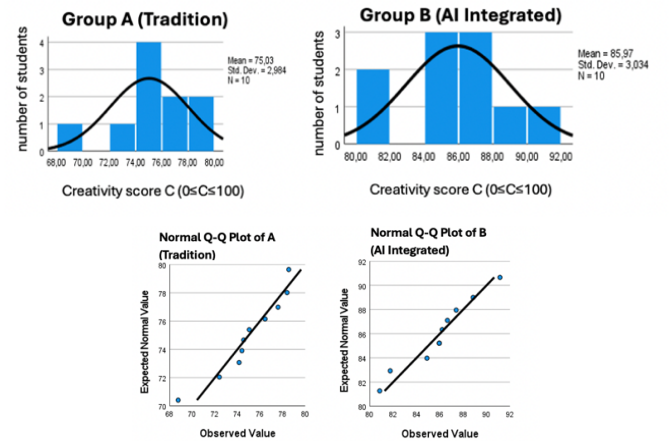


Fig 3. Normality assumptions of the dataset were assessed through Q-Q plots and histograms

Given the small sample size (N = 10), data suitability for group comparison was examined using visual diagnostics, including Q-Q plots and Histogram (Fig 3). The absence of extreme outliers and the approximately symmetric score distributions support the validity of comparative analysis between the two groups.

Following the success of the pilot, the second phase of the study was conducted with the administration of the survey on 60 students who followed the entire AI-supported curriculum in the 2024.2 semester. The survey tool consisted of nine questions, divided into three core conception areas (Table V), with three

open-ended questions, allowing the quantitative treatment of the answers with the help of the five-point likert-like scale for the open-ended answers: 1-None; 2-Slight; 3-Moderate; 4-Relatively Good; 5-Very Good

TABLE V . QUESTIONS FROM THE SURVEY THAT EVALUATE STUDENTS' SATISFACTION WITH THE APPLICATION OF THE MIDJOURNEY ARTIFICIAL INTELLIGENCE TOOL

ID	Content Domain	Survey Question
1	Motivation	Was AI involved in you participating in the class learning activities?
2		Did the AI system promote interaction with other students in the class?
3		Did the motivation or feelings of collective responsibility about homework completion come from AI?
4	Ease of Use	Did specialized training or guidance help in effectively utilizing the AI tool?
5		Are the operational procedures for the AI application straightforward?
6		Will the application work well on different technology gadgets?
7	Positive Impact	How much do you prefer group work with the support of the AI application?
8		What is your emotional state/response/experience working on the AI application?
9		How was the AI application beneficial in aiding your learning process?

TABLE VI . ASSESSMENT OF POSITIVE IMPACT LEVELS IN THE AI-INTEGRATED TEACHING MODEL

ID	Survey Question	Assessment Level (%)				
		1	2	3	4	5
1	Level of preference for group work on the AI app?	-	-	5	6	89
2	Emotional experience while working on the AI apps?	-	-	7	9	84
3	How did the AI application help you with learning?	-	-	9	24	67

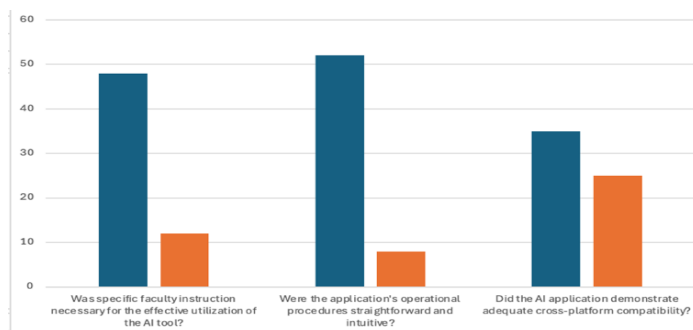


Fig. 4. Study results showing that the Midjourney AI app is easy to use in the learning process.

The collective outcomes from the research clearly show that over 89% of the students surveyed found the application of AI technology in the DT process to be of "good" to "very good" significant value (see Table VI).

B. Discussion of Research Outcomes

The collective result, represented in Fig. 4, clearly justifies the fact that the incorporation of AI technology inside the Design Thinking curriculum is highly beneficial for the learners. The AI-enabled environment clearly facilitated the collaborative process among the learners and highly boosted their intrinsic

motivational levels throughout the Design Thinking activity [11][20].

The proposed model not only conserved the time of the learners, enabling them to accomplish their tasks efficiently, but also boosted collective working, responsibility, and the learners' learning passion simultaneously. The applications of AI technology worked miraculously well in the ideation & screening stage of the process as well, because they added substantially to the learners' sustained interactions with each other, known as peer-to-peer interactions, along with interactions with their faculty members, specifically in the refinement & validation stage of the activity.

Despite promising outcomes, the deployment of AI in Design Thinking is not without systemic friction. We identified three primary hurdles: the preparedness of faculty, infrastructural deficits, and the integrity of assessment. Faculty members face a steep learning curve; mere familiarity is insufficient—they require fluency in managing AI-driven workflows. Institutions must respond with targeted training programs that go beyond basics to include practical, scenario-based classroom management. Infrastructurally, the lack of universal access to premium software and computing power remains a barrier, necessitating aggressive pursuit of educational

licensing partnerships. Perhaps most critical is the challenge to assessment integrity. As AI tools obscure the origin of work, distinguishing human effort from machine output becomes difficult. Consequently, a shift towards 'process-based' or hybrid assessment is essential, allowing instructors to evaluate the evolution of an idea rather than just the final, polished product.

V. CONCLUSION AND FUTURE WORK

The integration of Artificial Intelligence (AI) into Design Thinking (DT) curricula has garnered positive student reception, driven by a shift toward more dynamic pedagogical interactions. The performance gap observed in the computer mouse design experiment was substantial: the AI-supported group (Group B) did not just outperform the control group (Group A)—they eclipsed them. Group B achieved superior creativity scores (86 vs. 75) and produced 2.4 times the volume of concepts, all while slashing ideation time from 156 minutes down to a mere 20.

This data suggests AI acts as a multiplier for design exploration, offering a broader and more flexible solution space. Expanding the scope to 60 students in a subsequent semester, we found that 84% recognized the practical value of these tools, and over 90% cited significant workflow optimization. While the AI-integrated DT model stands out as a potent framework for creativity, it brings a caveat: instructors must establish clear guardrails to mitigate over-reliance. Assessments must be recalibrated to explicitly credit students' reasoning rather than just the final output. Our ongoing work will extend this inquiry, testing the model's efficacy across diverse art and design contexts.

VI. ETHICAL & COMPLIANCE STATEMENT

This study was conducted in strict accordance with established ethical guidelines for educational research. Formal approval was secured from HUST to ensure full compliance with standards regarding human participants. Student participation in the AID course (Semesters 2024.1 and 2024.2) was entirely voluntary, predicated on a transparent disclosure of the research scope and methodology. We obtained written informed consent from all participants, with the explicit guarantee that withdrawal at any stage would incur no academic or professional penalty. Furthermore, all digital training activities were designed to align with HUST's protocols on academic integrity and sustainability.

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